

Third Semester B. Tech. (Electronics and Communication Engineering) Examination**SIGNALS AND SYSTEMS**

Time : 3 Hours]

[Max. Marks : 50

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Marks as indicated.
- (3) Neatness carries due weightage.

1. (a) Sketch and estimate the nature of following signals :

(i) $y(t) = [u(t) + r(t - 1) - 2u(t - 3)] \cdot u(-t + 5)$.

(ii) $y(n) = 2x(n) + \delta(n)$

if $x(n) = 2^{-n}$ for $-2 \leq n \leq 2$.

5(CO1)

- (b) Consider the system :

$$y(t) = x^2(t - t_0) + 2$$

Where t_0 is a positive constant.

Determine whether the system is –

(i) Linear,

(ii) Time variant,

(iii) Stable,

(iv) Causal.

5(CO1)

2. (a) Judge whether the given systems are causal and stable –

(i) $h(n) = 5^n u(3 - n)$

(ii) $h(n) = e^{2n} u(n - 1)$

5(CO5)

(b) Find the convolution of following signals :

$$(i) \quad x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$(ii) \quad h(n) = u(n) \quad 5(CO2)$$

3. (a) Illustrate the steady step response of the following system to unit step excitation :

$$H(s) = \frac{s + 1}{s^2 + 3s + 2} \quad 5(CO3)$$

(b) Use convolution theorem of Laplace transform to find :

$$y(t) = x_1(t) * x_2(t)$$

where $x_1(t)$ and $x_2(t)$ are given below

$$x_1(t) = e^{-3t}u(t) \text{ and } x_2(t) = u(t - 2). \quad 5(CO3)$$

4. (a) Find the inverse z-transform of

$$X(z) = \frac{(2 + z^{-2})^2}{\left(1 - \frac{1}{2} z^{-1}\right) \left(1 + \frac{3}{4} z^{-1}\right)} \quad 5(CO4)$$

(b) Determine the value of the signal at $n = 0$ and $n = \infty$ if

$$X(z) = \frac{2z^2 + 0.25}{(z + 0.25)(z - 1)}. \quad 5(CO4)$$

5. (a) Justify the mapping between S-plane and Z-plane with the help of plotting of both planes. 5(CO3,4)

(b) Find the difference equation satisfying the input and output of an LTI system with transfer function :

$$H(z) = \frac{(1 + z^{-1})^2}{\left(1 - \frac{1}{2} z^{-1}\right) \left(1 + \frac{3}{4} z^{-1}\right)} \quad 5(CO4)$$

